WW-P-491C
August 25, 1978
SUPERSEDING
Fed. Spec. WW-P-491B
May 1, 1967

FEDERAL SPECIFICATION

PIPE FITTINGS; CAST-IRON, (THREADED): DRAINAGE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cast-iron drainage pipe fittings with threaded ends.

1.2 Classification.

1.2.1 Types and styles. Cast-iron threaded drainage fittings shall be of the following types and styles specified by the Military Standard part number (see figure 1 and 6.2):

Type I - Uncoated.
Type II - Coated (painted).
Type III - Zinc coated (galvanized).

Style 1 - Regular threaded end.
Style 2 - Reduced size inlet threaded end.
Style 3 - Bushed threaded end.

1.2.2 Configuration and Size. Types of fittings covered by MS90522 through MS90528 shall be of the configuration and size specified by the Military Standard part number (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

FED. STD. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards

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and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, D.C., Chicago, Kansas City, MO., Fort Worth, Houston, Denver, San Francisco, Los Angeles, Atlanta, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-V-3 - Valve, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS90522 - Elbow, Cast-Iron, Threaded Drainage.
MS90523 - Tees, Y-Branches, Crosses, Cast Iron Threaded Drainage.
MS90524 - Y-Branches, Reducing, Cast-Iron Threaded drainage.
MS90525 - Y-Branches, 45 Degrees, Reducing, Cast-Iron Threaded Drainage.
MS90526 - Couplings, Increases, Offsets Cast-Iron Threaded Drainage.
MS90527 - Y-Branch, Tucker and Roof Connections, Cast-Iron Threaded Drainage.
MS90528 - Trap, P-Bath and Running, Cast Iron Theaded Drainage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American National Standards Institute (ANSI):

B16.12 - Cast Iron Threaded Drainage Fittings.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM):

A126 - Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
A153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
A197 - Cupola Malleable Iron.
A536 - Ductile Iron Castings.
E351 - Chemical Analysis of Cast Iron - All Types.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Fittings shall be made from gray iron conforming to ASTM A126. The manufacturer may supply fittings made from ductile iron conforming to ASTM A536 or A197.

3.2 Coating.

3.2.1 Type I. Type I fittings shall be uncoated.

3.2.2 Type II. Type II fittings shall be painted with a thin, hard, adherent, rust-protective black lacquer or black enamel. All surfaces of the fittings, except threads, shall be coated.

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3.2.3 Type III. Type III fittings shall be coated by the hot-dip process and so treated as to be thoroughly coated both inside and outside with a smooth, bright, adherent surface of zinc conforming to class A of ASTM A153. The coated fittings shall be free of blisters, dross, dirt, bare spots, rust, and coating defects. The fittings shall be coated before threading.

3.3 Threads. All fittings shall be threaded in accordance with ANSI B16.12.

3.4 Dimensional tolerances. Tolerances on dimensions for all fittings, sizes 1-1/4 inch through 8 inches, shall be in accordance with ANSI B16.12, table I.

3.5 Design.

3.5.1 Threaded ends for styles 1, 2, and 3. The female threaded ends of style 1, 2, and 3 fittings (see figure 1), shall conform to the dimensions specified in ANSI B16.12

3.5.2 Ends. The ends shall be threaded female and the style (figure 1) of the ends shall be as specified by the Military Standard part number (see 6.2). The ends shall conform to the dimensions of the appropriate table of ANSI B16.12 except that inlet connections which are smaller than the outlet connections, must be type 3 (style 2) or type 4 (style 5) as specified. (NOTE: Style 1 of figure 1 includes type 1 and type 2 of ANSI B16.12-77, table 3).

3.6 Pressure test. When specified (see 6.2 and 6.3), fittings covered by this specification shall not leak when subjected to a minimum 60 pounds per square inch (psi), air pressure in accordance with 4.4.1.

3.7 Identification marking. The manufacturer's name, initial, or other designation by which he can be easily identified, shall be legibly cast on each fitting.

3.8 Workmanship. The quality of workmanship shall be such as to produce fittings that are in accordance with the requirements of this specification. The fittings shall be free from sand and scale.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Sampling for examination. A random sample of fittings shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The Acceptable Quality Level (AQL) shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.2.2 Sampling for tests.

4.2.2.1 Sampling for air pressure test. When specified, a random sample of fittings shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be 4.0 percent defective.

4.2.2.2 Sampling for chemical analysis. Unless otherwise specified (see 6.2), the manufacturer shall submit a ladle analysis as proof that the fittings submitted meet the requirements of 3.1.

4.2.2.2.1 Sample selection. When specified (see 6.2), two 2-ounce samples shall be taken from each lot and shall be subjected to chemical analysis specified in 4.4.2.

4.3 Examination. Each of the sample fittings selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table I.

Table I. Classification of defects

Catagories	Defects
Major:	
101	Kind of fitting not as specified.
102	Size not as specified.
103	Threads not as specified.
104	Surface coating not as specified.
105	Surface coating not uniform.
106	Cracks or blow holes present.
107	Dimensions exceeding tolerances.
Minor:	
201	Identification marking not as specified.
202	Sand or scale on surface.

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4.4 Test methods.

4.4.1 Air pressure. When specified (see 6.2 and 6.3), the sample fittings selected in accordance with 4.2.2.1 shall be subjected to at least 60 psi air pressure while under water. Any leakage of air shall be cause for rejection.

4.4.2 Chemical composition. When specified (see 6.2), samples selected in accordance with 4.2.2.2.1 shall be analyzed in accordance with ASTM E351.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the fittings shall be inspected to determine conformance with applicable requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The fittings shall be preserved and packaged in accordance with the applicable level A requirements of MIL-V-3.

5.1.2 Level C. The fittings shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The fittings shall be packed in accordance with the applicable level A requirements of MIL-V-3.

5.2.2 Level B. The fittings shall be packed in accordance with the applicable level B requirements of MIL-V-3.

5.2.3 Level C. The complete fittings shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

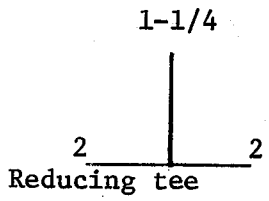
6. NOTES

6.1 Intended use. Fittings covered by this specification are intended for general drainage applications.

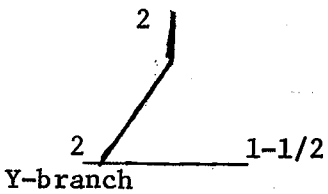
6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of finish required (see 1.2.1).
(c) Military Standard part number required (see 1.2.2 and 3.5.2).
(d) When pressure test is required (see 3.6, 4.4.1, and 6.3).
(e) When chemical analysis is required (see 4.2.2.2, 4.2.2.2.1, and 4.4.2).
(f) Selection of applicable level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(g) Marking required (see 5.3).

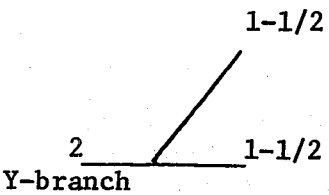
6.2.1 Ordering recommendations. In ordering reducing fittings, the kinds of fittings and sizes of the connecting ends shall be specified. In giving the sizes of reducing tees and Y-branches, the larger run is given first, the smaller run next, and the inlet or branch last. The following diagram illustrates the above method:



Reducing tee (2 X 2 X 1-1/4)



Y-branch (2 X 1-1/2 X 2 reducing 90° short)

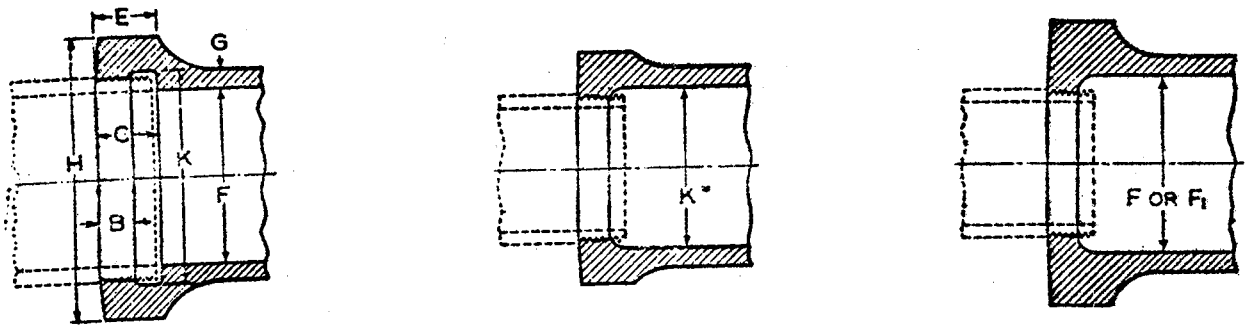


Y-branch (2 X 1-1/2 X 1-1/2 reducing 45°)

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6.3 Pressure test. Drainage fittings intended for low pressure service are not required to be pressure tested.

6.4 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.



Style 1
Regular

Style 2
Reduced size inlet

Style 3
Bushed

Figure 1. Threaded ends of cast iron drainage fittings

Military custodians:

Army - ME
Navy - YD
Air Force - 99

Review activity:

DLA - CS

Civil Agency Coordinating Activities:

GSA - FSS
VA - DMS

Preparing activity:

Navy - YD

Project No. 4730-0496

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